

Class 12 Biology: High-Yield Diagram Checklist

Chapter 9: Biotechnology – Principles and Processes (NCERT Page-wise Guide)

Instructions for Students: Practice sketching and labelling all biotechnology flowcharts, vectors, and apparatus neatly with pencil. Ensure all restriction sites, antibiotic markers, and PCR temperatures/enzymes are precisely indicated in BLOCK letters.

Fig No.	Diagram / Flowchart Title	NCERT	Priority	Key Components / Labelling Points
Fig 9.4	E. coli Cloning Vector pBR322	Pg 169	★★★ Top Priority	Markers: amp ^R (Pvu I, Pst I), tet ^R (BamH I, Sal I); Sites: EcoR I, Cla I, Hind III; ori, rop (Pvu II)
Fig 9.6	Polymerase Chain Reaction (PCR)	Pg 172	★★★ Top Priority	3 Steps: (1) Denaturation (Heat), (2) Annealing (Primers), (3) Extension (Taq polymerase + dNTPs) → ~1 billion times
Fig 9.7 (a, b)	Bioreactor Types (Simple vs Sparged Stirred-Tank)	Pg 174	★★★ Top Priority	Motor, Impeller blades, Foam breaker, pH/Temp control, Steam/Air ports, Sparger (bubbles for surface area)
Fig 9.1	Action of Restriction Enzyme (EcoRI) & Sticky Ends	Pg 166	★★★ Top Priority	Palindromic sequence 5'-GAATTC-3', staggered cuts between G & A, sticky ends, recombinant DNA with DNA ligase
Fig 9.2	Recombinant DNA Technology Process Flow	Pg 167	★★☆ High Priority	Foreign DNA + Vector DNA → Cut with same RE → Ligase → rDNA → E. coli Host → Transformation & Cell Division
Fig 9.3	Agarose Gel Electrophoresis Setup & DNA Bands	Pg 168	★★☆ High Priority	Wells (-ve cathode), Migration towards +ve anode, Largest to smallest bands, Ethidium bromide + UV light, Elution
Fig 9.5	DNA Spooling	Pg 171	★★☆ Moderate	Precipitation of purified DNA with chilled ethanol into visible fine threads; spooling collection

KEY BOARD EXAM DRAWING & CONCEPTUAL GUIDELINES

- **pBR322 Vector Accuracy (Fig 9.4):** Ensure restriction sites are placed in the correct antibiotic resistance domains (BamH I & Sal I in *tet^R*; Pst I & Pvu I in *amp^R*; Pvu II in *rop*).
- **PCR 3 Steps (Fig 9.6):** Clearly depict double-stranded DNA separation (Denaturation), oligonucleotide primer attachment (Annealing), and thermostable *Taq polymerase* extension.
- **Bioreactor Contrast (Fig 9.7):** Show flat-bladed impeller and ports in simple stirred-tank; emphasize sparged bubbles increasing oxygen transfer area in sparged reactor.